

CRYSTAL CALIBRATION SALES AND SERVICE CO., LTD.

45/48 Soi Salathammasop31, Salathammasop Rd.,

Salathammasop, Thawewatthana, Bangkok 10170 Thailand

Tel : 0-2408-8474-5 Fax : 0-2408-8477 Email : info@crystalcal.com www.crystalcal.com



CERTIFICATE OF CALIBRATION

Certificate No. : 25-1390-141

Issue Date : 5 November 2025

Work Order No. : 25/1390

Customer Name : กลุ่มงานเทคนิคการแพทย์ โรงพยาบาลพระจอมเกล้า จังหวัดเพชรบุรี
53 ถ.รถไฟ ต.คลองกระแซง
อ.เมือง จ.เพชรบุรี 76000

Date of Received : 29 October 2025

Date of Calibration : 3 November 2025

Instrument Details : Description : Digital Thermo hygrometer
Manufacturer : N/A
Model : N/A
Serial No. : N/A
ID No. : B00987
Location : Humidity and Temperature Laboratory

Calibration Method : This instrument was calibrated by comparison of indication with Standard Chilled Mirror Hygrometer and Standard Thermometer into Temperature and Humidity Chamber controller according to calibration procedure no. CWI-H-01

Environmental Condition

Temperature : Laboratory Control at $23^{\circ}\text{C} \pm 3^{\circ}\text{C}$

Humidity : Laboratory Control at $55\%\text{RH} \pm 20\%\text{RH}$

Traceability of Measurement

: This certificate of calibration documents the traceability to national standard, which realize the unit of measurement according to the International system of Units (SI) and The temperature scale in use at this laboratory is The International Temperature scale of 1990.

Calibrated by : Miss Phiraya Prawabut
Calibration Engineer

Approved by : 
(Mr. Anuwat Yaklermjit)
Laboratory Manager

This certificate may not be reproduced other than in full except with the prior written approval of Crystal Calibration Sales and Service co., Ltd.

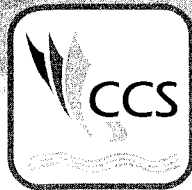
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Details of Calibration

1. Reference Standards Instrument

Instrument	Serial No. / ID no.	Certification	Due Date
1.1 Chilled Mirror Hygrometer	157152 / CMH-01	TH-0050-25	22 August 2027
1.2 Digital Thermometer with RTD	15000016 / RTD-11	25-1153-001	08 September 2026

2. Certificate traceable : This certificate traceable to The International System of Unit refer to
No. 1.1 National Institute of Metrology (Thailand), NAC Calibration no. 0144
No. 1.2 Crystal Calibration Sales and Service Co., Ltd. , NAC Calibration No. 0260

3. Condition of item : Used

4. Calibration location : Permanent

Result of Calibration

1. Temperature Measurement : Without Adjustment Resolution of UUC : 0.10 °C

Calibration Point (°C)	Average Standard Reading (°C)	UUC Reading (°C)	Correction (°C)	Uncertainty ± (°C)
23	23.062	22.40	+ 0.662	0.30
25	25.050	24.80	+ 0.250	0.30

2. Humidity Measurement : Without Adjustment Resolution of UUC : 0.10 %RH

Calibration Point (%RH)	Calculated Standard Reading (%RH)	UUC Reading (%RH)	Correction (%RH)	Uncertainty ± (%RH)
50	49.93	60.90	- 10.97	1.2
65	64.96	76.00	- 11.04	1.4

- Note :** 1. Process calibration humidity measurement Reference temperature control at 25°C
2. Calculated STD humidity refer to dew-point temperature and convert to humidity by magnus's Equation
3. Calibrate items it good condition and this report customer request and accepted in certificate

This result of calibration was found accurate as shown on date and place of calibration only.

The reported uncertainty of measurement was based on a standard uncertainty multiplied by a coverage factor $k=2$ providing a level of confidence of approximately 95%.



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Statements of conformity report

Refer to Certificate No. : 25-1390-141

Issue Date : 5 November 2025

Work Order No. : 25/1390

Detail of Equipment

Description : Digital Thermo hygrometer

Manufacturer : N/A

Serial no. : N/A

Model : N/A

ID No. : B00987

Result of Calibration

This result of calibration : Without Adjustment

MPE part Temp. 1 °C

Resolution : 0.1 °C

Calibration point	STD value	UUC reading	Correction	Uncertainty	Uncertainty + Absolute correc.	Judgment
23	23.062	22.40	0.662	0.30	0.962	Pass
25	25.050	24.80	0.250	0.30	0.550	Pass

MPE part Humid 5 %RH

Resolution : 0.1 %RH

Calibration point	STD value	UUC reading	Correction	Uncertainty	Uncertainty + Absolute correc.	Judgment
50	49.93	60.90	-10.97	1.2	12.17	Fail
65	64.96	76.00	-11.04	1.4	12.44	Fail

The conformity certificate documents validity following ISO/IEC Guide 98-4 : Role of measurement uncertainty in conformity assessment based on statement with guard band refer to specification tolerance limit of customer consider expanded measurement uncertainty (k=95%)

The tolerance and decision rules ;

Consideration Guidelines = Measurement uncertainty + Absolute error ; $\leq$ MPE is pass , $>$ MPE is Fail

Statements of conformity decision by :

(Mr. Anuwat Yaklermjit)

Laboratory Manager

This document is a guideline for interpreting calibration report for customers only is not part documentation ISO/IEC 17025:2017

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